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[22ME43]

FOURTH SEMESTER B.Tech EXAMINATION, MAY - 2024

PRECISION ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 x 02=10**		
1.	**a.**	Define precision in measurement.	L1	CO1	**02**
	b.	State and explain main function of a comparator.	L1	CO2	**02**
	c.	Describe the basic principle behind interferometry used for measurement applications.	L2	CO3	**02**
	d.	Distinguish accuracy and precision in measurement.	L2	CO4	**02**
	e.	List the difference between hysteresis and repeatability in a measurement system.	L1	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Discuss the importance of standards in manufacturing and engineering. Provide two examples of prominent standards organizations.	L2	CO1	**05**
		OR			
	b.	Illustrate three factors to consider when selecting a measuring instrument for a specific task.	L2	CO1	**05**
3.	**a.**	Discriminate between resolution and accuracy in measurement. Provide an example for each.	L3	CO1	**05**
		OR			
	b.	Explain the concepts of flatness, straightness, and squareness with the help of appropriate measuring tools.	L2	CO1	**05**
4.	**a.**	Describe any two types of comparators and their applications in machining and inspection.	L2	CO2	**05**
		OR			
	b.	Discuss the use of a sine bar? How is it used to measure angles?	L2	CO2	**05**
5.	**a.**	Explain the applications of autocollimators in precision measurement tasks.	L2	CO3	**05**
		OR			
	b.	Explain how the choice of measurement method (e.g., micrometer vs. two-wire method) can influence the accuracy and ease of measurement.	L2	CO3	**05**

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6.	a.	List out the main components of a generalized measurement system and discuss the function of each component.	L2	CO4	05
		OR			
	b.	Illustrate the different types of errors encountered in measurements and provide examples of each type.	L2	CO4	05
7.	a.	Describe the concept of calibration and explain its importance in maintaining the accuracy of measurement systems.	L2	CO5	05
		OR			
	b.	Explain two methods for measuring temperature and discuss the advantages and limitations of each method.	L2	CO5	05
		PART-C			**06 X 10=60**
8.	a.	Discuss any two applications of interferometry in precision measurement. Explain the specific principles involved in each application.	L2	CO3	10
		OR			
	b.	Identify and explain two limitations associated with using interferometry for measurement tasks.	L1	CO3	10
9.	a.	Compare and contrast the two-wire and three-wire methods for measuring screw thread diameters. Discuss the factors to be considered when choosing between these methods.	L3	CO3	10
		OR			
	b.	Explain how a toolmaker's microscope can be used for inspecting and measuring screw thread features. Describe the advantages of using a microscope for this purpose compared to micrometers.	L1	CO3	10
10.	a.	Describe the main components of a generalized measurement system and explain the function of each component in converting a physical quantity into a readable output.	L2	CO4	10
		OR			
	b.	Discuss the concept of errors in measurement and differentiate between systematic and random errors.	L2	CO4	10
11.	a.	Explain the concept of sensor sensitivity and its importance in measurement systems. Discuss factors that can influence the sensitivity of a sensor and how to improve it.	L2	CO5	10
		OR			

	b.	Discuss the operating principles, advantages, and limitations of any two types of sensors used in measurement applications.	L3	CO5	**10**
12.	**a.**	Briefly explain the temperature measurement using resistance thermometer and thermocouples with their merits and demerits.	L4	CO5	**10**
		OR			
	b.	Discuss the principle behind pyrometry and differentiate between optical and radiation pyrometers.	L2	CO5	**10**
13.	**a.**	Explain the working principle of a strain gauge and how it converts mechanical strain into an electrical signal. Discuss the factors that influence the sensitivity of a strain gauge.	L2	CO5	**10**
		OR			
	b.	Describe the functionalities of a Universal Measuring Machine (UMM) and explain its advantages for various measurement tasks compared to simpler tools like calipers or rulers.	L2	CO5	**10**